TARGET-FREE PRIME-POWER REDUCTION, SHORT-ARC CUBIC OBSTRUCTIONS, AND INVERSE-FROBENIUS CLASSIFICATIONS FOR AGRAWAL'S CONJECTURE

ARGUS OPEN RESEARCH PROJECT

Abstract. Agrawal's conjecture predicts that if $n \geq 2$ is coprime to r and $F_n(X) = (X-1)^n - X^n + 1 \equiv 0 \pmod{(n, X^r - 1)}$, then either n is prime or $n^2 \equiv 1 \pmod{r}$. We establish three unconditional structural results on the cyclotomic congruences governing hypothetical counterexamples. First, we prove a target-free prime-power reduction: $F_n(X) \equiv 0 \pmod{(n, X^r - 1)}$ if and only if $F_n(X) \equiv 0 \pmod{(\mathrm{rad}(n), X^r - 1)}$ and every prime power $p^e \parallel n$ satisfies $e \leq w(p, r)$. Second, for odd primes p primitive modulo a prime $r \geq 37$ with $p^2 \mid n$, we classify short-arc rational cross-ratios $R_{x,y,z}(T) = \frac{(1-T^{x+y})(1-T^{y+z})}{(1-T^{x+y+z})(1-T^{y})}$ in the finite logarithm $L_p(X)$ to prove the explicit cubic lower bound $p \geq \max\{r^2 + 2,\ 1 + N(\lfloor (r-1)/4 \rfloor)\}$ where $N(M) = \frac{1}{2}(\binom{M}{3} + \lfloor (M-1)^2/4 \rfloor) \sim \frac{1}{768} r^3$, whence $n \geq p^2 \gg r^6$. Third, in the index-two cyclotomic quotient $\mathbb{F}_p[X]/\Phi_r(X)$ ($r \equiv 3 \pmod 4$, $\mathrm{ord}_r(p) = (r-1)/2$), we classify all inverse-Frobenius solutions $(X-1)^k \equiv X^{-p^j} - 1$ via the norm identity $N(2 - X - X^{-1}) = r$ and prove that admissible primes have positive natural density for $r = 7$.

1. Main Mathematical Results

Theorem 1.1 (Prime-power reduction, cubic obstruction, and inverse-Frobenius classification)**.** *Let* $F_n(X) = (X-1)^n - X^n + 1$.

(1) *For every* $n \geq 2$ *and* $r \geq 2$ *with* $\gcd(n, r) = 1$, $F_n(X) \equiv 0 \pmod{(n, X^r - 1)}$ *holds if and only if* $F_n(X) \equiv 0 \pmod{(\mathrm{rad}(n), X^r - 1)}$ *and* $F_p(X) \equiv 0 \pmod{(p^e, X^r - 1)}$ *for every* $p^e \parallel n$.

(2) *For every prime* $r \geq 37$ *and odd prime* p *with* $\mathrm{ord}_r(p) = r - 1$, *if* $F_p(X) \equiv 0 \pmod{(p^2, X^r - 1)}$, *then with* $M = \lfloor (r-1)/4 \rfloor$,

$$p \ \geq \ \max\left\{r^2 + 2,\ 1 + \frac{1}{2}\left(\binom{M}{3} + \left\lfloor \frac{(M-1)^2}{4} \right\rfloor\right)\right\},$$

and any solution n *of* $F_n \equiv 0 \pmod{(n, X^r - 1)}$ *divisible by* p^2 *satisfies* $n \geq p^2$.

(3) *For prime* $r \equiv 3 \pmod 4$ *and odd prime* p *with* $\mathrm{ord}_r(p) = (r-1)/2$, *the equation* $(X-1)^k = X^{-p^j} - 1$ *in* $\mathbb{F}_p[X]/\Phi_r(X)$ *is solvable if and only if* $\gcd(r, m) = 1$ *and* $v_2(\mathrm{ord}_{\mathbb{F}_p^\times}(r)) \neq 1$.

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 9400114 in the domain of computational and algebraic number theory adheres to a strict separation between the exact statement proved and adjacent formulations with stronger or weaker hypotheses:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, number-field, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, finite-field, or symbolic arithmetic without floating-point approximation, and independently re-verified from standalone certificates.

(c) *Resolved scope:* Target-free radical/multiplicity equivalence for F_n = 0 mod (n, X^r-1), explicit cubic primitive-prime bound p >= max(r^2+2, 1+N(floor((r-1)/4))) r^3/384, and complete inverse-Frobenius cyclotomic norm classification.

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3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that remain open beyond the scope settled above:

- Unrestricted squarefree composite counterexamples to Agrawal's conjecture

References

[1] Primary problem list, literature attributions, and exact verification certificates for Result 9400114 (computational and algebraic number theory), archived in the accompanying repository package `results/9400114/`.